

Sreya Vadapalli

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Professional Summary

Biomedical and public health data scientist with 3+ years of experience applying machine learning, artificial intelligence, and statistical modeling across epidemiology, precision medicine, and behavioral health research.

Education

Rutgers University (2023)

Master of Business and Science in Computer Science
(Concentration: Analytics), GPA: 3.7

New Brunswick, New Jersey.

C.B.I.T, Osmania University (2020)

Bachelor of Technology, Biotechnology, GPA: 8.41/10 (3.9)

Hyderabad, India.

Professional Experience

Rogers Behavioral Health, Milwaukee

Aug 2023 - June 2026

Research Associate II

- Quantified a **\$4.2B** annual healthcare burden across **108** rare genetic disorders with primary neuropsychiatric symptoms and a **7.7**-year diagnostic delay, building the case for earlier genetic testing.
- Analyzed clinical outcomes (quality of life, depression, anxiety, OCD severity) and medication utilization (polypharmacy) across matched cohorts at admission and discharge in retrospective studies of OCD and autism spectrum disorder.
- Anxiolytic Management Quality Improvement Project: Evaluated anxiolytic management in **242** eating-disorder patients using retrospective EHR data from **2018–2024**; matched **121** anxiolytic-treated patients with **121** controls on age, sex, race, and baseline EDE-Q scores using MatchIt in R. Compared changes in EDE-Q, YBC-EDS, QIDS, Y-BOCS, exposure-hierarchy completion, length of stay, and readmission across **4** levels of care; anxiolytic-treated patients demonstrated **27–40%** improvement across symptom measures.
- Developed an integrated data pipeline to process, extract, and investigate patient data.

Rutgers University, New Jersey (during MBS degree)

Robert Wood Johnson Medical School, NJ

Aug 2022 - May 2023

Research Assistant

- Implemented ensemble decision forests model on EEG data to study post-ictogenesis achieving **91.4%** accuracy and **0.97** AUC.
- Performed neural networks for auto-segmentation of pre-ictal and ictal seizure phases in epilepsy.
- Analyzed Racine score associations with the number of seizures and days using statistical methodologies.

Rutgers MBS Externship Exchange, NJ

Sep 2022 - Dec 2022

Data Science Extern (COVID-19 Epidemiology Bootcamp)

- Assembled and curated multi-source COVID-19 data (cases, fatalities, NWSS wastewater, county/state) and developed predictive ML models (neural networks, logistic regression, and XGBoost), achieving **77%** accuracy in state-level variant prediction with XGBoost.

Rutgers Institute of Health, Health care Policy, and Aging Research, NJ

Nov 2021 - May 2022

Data Science Research Assistant and Teaching Assistant

- Reviewed and analyzed **32** AI, ML approaches applied to whole-genome, whole-exome sequencing, and gene expression data; identified widely adopted predictive algorithms for precision medicine across multiple diseases.
- Contributed to RNA-seq-driven transcriptomic analysis to identify **3** highly expressed AF-associated genes (*PF4*, *PPBP*, *MYL4*) and their phenotypic associations across race and gender.
- Led weekly discussions and held office hours for **20** undergraduate students enrolled in Byrne Seminar - Practicing Precision Medicine with Data Analysis.

Nizam's Institute of Medical Sciences

Medical Biotech and Statistics Intern

May 2019 - Jul 2019

- Investigated the association between Taq1A and Taq1B genetic polymorphisms and Parkinson's disease susceptibility in South Indian populations.
- Compared genotype and allele frequency distributions between Parkinson's disease cases and controls to identify statistically significant genetic differences.

Publications

- Artificial intelligence & machine learning approaches using gene expression and variant data for personalized medicine.(Oxford, Briefings in Bioinformatics)
- Artificial Intelligence, Healthcare, Clinical-Genomics, and Pharmacogenomics Approaches in Precision Medicine. (Frontiers, Genetics)
- RNA-seq-driven expression analysis to investigate cardiovascular disease genes with associated phenotypes among Atrial Fibrillation patients. (Wiley, Clinical and Translational Medicine)
- Polypharmacy and pharmacogenomics in high-acuity behavioral health care for autism spectrum disorder: a retrospective study. (Springer, CAPMH)
- Pharmacogenomic-guided prescribing and polypharmacy across age groups in obsessive-compulsive disorder: a retrospective study. (Springer, BMC)
- Bridging psychiatry and rare genetic diseases: a scoping review of therapeutic strategies and diagnostic delay paired with healthcare

economic burden analysis. (Springer, Orphanet Journal of Rare Diseases)

7. Multi-Frequency Interpolation X-talk Removal Algorithm: Enabling Combinations of Concurrent Optogenetics and Lock-in Amplification Fiber Photometry via Removal of Optogenetic Stimulation Crosstalk. (ACS, Chemical Neuroscience)

Conference Abstracts

- P342: Utilization of pharmacogenomics in a child and adolescent autism spectrum disorder depression and anxiety program: A retrospective study (ACMG, Genetics in Medicine)
- 3.38 Utilization of Pharmacogenomics for ASD: A Retrospective Study (Journal of American Academy of Child and Adolescent Psychiatry)

Preprints

- Anxiolytic management in eating disorder patients receiving cognitive behavioral therapy: A quality improvement brief report. (medRxiv, Preprint)

Presentations and Talks

- Presented a poster titled Tailoring Exposure and Response Prevention to Treat Disgust OCD at the International OCD Foundation conference in Chicago, **2025**.
- Invited talk / Webinar : Clean Sheets or Muddy Boots: A Discussion of Curated and Field Data Sets (Midwest Big Data Innovation Hub Data Science Student Groups Webinar)

Skills

Programming: Python (pandas, NumPy, scikit-learn, PyTorch), R, SQL, Tableau, MATLAB

Machine Learning: Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Deep Learning

Gen AI: LLMs, NLP, Transformers, RAG

Genomics: NGS Data Processing, WES, Variant Calling (GATK), Alignment (BWA)

Academic Projects

LitLens

- Built a biomedical RAG app using React, FastAPI, Claude API, SPECTER embeddings, and ChromaDB to parse, retrieve, and summarize research papers.
- Designed an evaluation framework for retrieval and generation quality using a hand-labeled gold set, including Hit@k, MRR, and claim-grounding checks.
- Improved retrieval precision through source-scoped filtering, raising Hit@5 from **0.55** to **0.75**.

Life Expectancy Prediction

- Fitted a Multiple Linear Regression model predicting life expectancy on WHO data (2000-2015), Ridge/Lasso regularization and country fixed effects improved R^2 from **0.81** to **0.94**.

Exploratory Analysis of Poverty in NYC

- Supervised ML implementation to predict poverty in the NYC population, resulting in an accuracy of **84%** by applying and comparing SMOTE, random oversampling, and random under-sampling to address class imbalance.

Awards and Achievements

- MBS Fellowship Aug 2022 & Jan 2023
- Chief Head, President, and Editorial in charge of Bioengineering and Biotechnology Club of C.B.I.T, 2019-2020.
- Presented a chapter presentation on projects done in C.B.I.T at the All-India EWB CHAPTERS MEET, Chennai, 2018.
- Core Committee Head - TEDx CBIT, 2018.
- Head of Logistics, Model United Nations CBIT 2018; Deputy Head of Logistics, Model United Nations CBIT, 2017.
- Silver Medal for Academic Excellence, 2016